



Use of Brain Computer Interface to Drive: Preliminary Results

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ABSTRACT

This paper reports on the implementation of a non-invasive electroencephalography-based brain-computer interface to control functions of a car in a driving simulator. The system is comprised of a Cleveland Medical Devices BioRadio 150 physiological signal recorder, a MATLAB-based BCI and an OKTAL SCANer advanced driving experience simulator.

The system utilizes steady-state visual-evoked potentials for the BCI paradigm, elicited by frequency-modulated high-power LEDs and recorded with the electrode placement of Oz-Fz with Fz as ground. A three-class online brain-computer interface was developed and interfaced with an advanced driving simulator to control functions of the car, including acceleration and steering.

The findings are mainly exploratory but provide an indication of the feasibility and challenges of brain-controlled on-road cars for the future, in addition to a safe, simulated BCI driving environment to use as a foundation for research into overcoming these challenges.

Categories and Subject Descriptors

H.5.2 [Information Interfaces and Presentation (e.g., HCI)]:
User Interfaces H.4 [Information Systems Applications]:
Miscellaneous

General Terms

Experimentation, Human Factors.

Keywords

Brain computer interface (BCI); car simulator;
electroencephalography (EEG); human computer interaction;
intelligent transport systems; steady state visual evoked potential
(SSVEP)

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1. INTRODUCTION

Recent progress in human-computer communication systems has shown that controlling objects with the mind is viable beyond the science fiction cliché. Advanced electrophysiological recording techniques now allow for monitoring of the physical processes within the brain, opening limitless opportunities in the field of brain-computer interfaces (BCIs).

Driven in-part by growing societal recognition for the needs of people with physical disabilities, brain-computer interfaces allow for tasks that would traditionally depend on peripheral nerves and muscles to instead be performed using brain activity [6]: several prototypes exist that enable users to control wheelchairs, keyboards, and computer games solely by means of cerebral activities sensed with electroencephalography (EEG) [4].

Particular patterns of brain activity in a user's EEG signals can be classified using signal processing and pattern recognition techniques, allowing a respective action to be performed based on the application.

The objective of this project was to create an EEG-based BCI using the available Cleveland Medical Devices BioRadio 150 and OKTAL SCANer advanced driving experience simulator installation at the Centre for Accident Research and Road Safety – Queensland, which features a six degree-of-freedom motion platform, 180° front field of view and three simulated mirrors [1].



Figure 1. CARRS-Q advanced driving simulator [1].

2. METHODOLOGY

An assessment of potential BCI paradigms was performed, leading to the decision to use an unconsciously-generated brain signal to classify the user's intention, as they require little training and use of the user's mental capacity. Of the unconscious BCI paradigms, steady-state visual evoked potential (SSVEP) was chosen as the stimulus, due to its requirement of only intermittent (peripheral) visual attention [3].

In this type of BCI, the user directs their visual attention to one of numerous uniquely frequency-modulated lights/images that represents their decision. When an SSVEP response is elicited by a stimulus at a particular frequency, there is a distinct peak in power spectral density at the stimulus frequency and its harmonics, in the EEG over the occipital lobes [6]. Matched filter detection can be implemented to determine the most significant stimulus frequency in the user's brain response and therefore decode the user's intent [2].

As these SSVEP response signals are generated automatically within the brain, there is often little specific training required for use; however there is a persistent need for external stimuli, which may not always be suitable or desirable for the application and/or user [4].

Electronic stimuli in the form of lights were chosen as the method of eliciting SSVEP, as they do not require a display/monitor and consequently the stimuli frequencies are not restricted by the displays' refresh rates [8].

Chang et al. [2] have shown that a remote-controlled car can be successfully controlled around an S-shaped track with an SSVEP BCI. Their implementation involved four flashing rectangles overlaid on video footage from the car's perspective on a computer monitor. Another relevant visual stimulation for a brain-controlled simulated car is discussed in Wang et al. [7]. This interface contained five flickering "buttons" that each consisted of 3x3 cm blue LEDs of varying frequency. These buttons were controls for an electrical control that represented forward, backward, left, right and stop. These examples support the decision to use SSVEP to control a car with EEG signals.

3. SSVEP IMPLEMENTATION

3.1 Light Stimuli

The steady-state visual-evoked response utilized for brain-computer interfaces requires flashing stimuli from external monitors or lights. Due to the intent to use the stimuli in a car application, light stimuli that can be positioned unobtrusively in the car were deemed appropriate. External light SSVEP stimuli were developed using three 17,000 mcd high-power LEDs placed inside a 5x5 cm box with a light-dispersing lens, frequency modulated by an Atmel ATmega168 microcontroller.

3.2 Electrode Placement

Multiple SSVEP trials were performed with different electrode placements, including monopolar and bipolar placements. Electrode positions of O1, Oz and O2 were tested with P1, P2, Cz, Fz left and right mastoid, and forehead as references. Greatest response with minimal setup was achieved with Oz and forehead for the monopolar channel and forehead as the ground. The slight sacrifice in performance in this setup with two electrodes, over the three-electrode setup of Oz and right mastoid with forehead as

ground, was deemed acceptable because of the accompanying reduction in system complexity.

3.3 Signal Processing

To decode the SSVEP stimulus in the user's visual focus and therefore the user's intent, the frequency of the SSVEP response was determined using a matched filter bank. The signals of the matched filter bank were complex sinusoids each with a frequency identical to the flashing frequency of one of the light stimuli. The matched filter was used to determine the strongest of the stimuli frequencies present in the user's brain signal and thereby classify the stimulus in focus [2].

3.4 Stimuli Frequencies

Different stimuli frequencies were experimented with to determine which three would reduce misclassification, and would therefore be most suitable for the online BCI interface with three SSVEP stimuli. When particular combinations of frequencies (e.g. 6, 7, and 8 Hz) were used as the stimuli frequencies, they were found to be more prone to misclassification as some frequencies (8 Hz, in this example) were found to be continually classified regardless of the stimulus in focus.

The combination of 5.783 Hz, 6.75 Hz and 8.65 Hz as the stimuli frequencies was found to yield the highest classification rate of approximately 90%, with the exception of the transition periods between changing stimulus focus. Misclassification occurred during these transition periods due to the larger window of time (approximately 3 seconds) that the analysis was performed on, in order to improve classification when only one stimulus was being focused on. This duration of time affected by this phenomenon can be reduced by reducing the length of the data buffer that is analyzed, however this may lead to reduced accuracy of the system when the user is not transitioning between stimuli, due to less SSVEP response being available in the data buffer.

4. DATA ACQUISITION

4.1 BioCapture

BioCapture is a software program provided with the wireless BioRadio 150 physiological monitor to capture and store EEG signal data. The software package allows real-time streaming of EEG data and this data can be viewed in a window containing a single or several channels, as specified by the user. BioCapture allows the user to configure and program the BioRadio sampling rate, number of enabled channels and the type of physiological signal collected. Data can be exported to ASCII format for analysis in other programs.

BioCapture was used for offline data collection of the EEG signal. 30-second trials were performed with the participant gazing at flashing stimuli for the SSVEP paradigm and the collected EEG data exported to comma-separated value files for analysis in MATLAB at a later time. Amplitudes in the frequency magnitude spectrum at the stimuli frequency were compared for various tests to improve SSVEP response and determine the effects of attributes such as electrode placement and stimuli characteristics.

4.2 MATLAB

MATLAB was selected as an appropriate development environment due to its rapid prototyping capability which aligned well with requirements for the project.

Data from the BioRadio 150 was collected periodically and captured in MATLAB as a stream of data points. A function was designed which initialized the communication with the BioRadio, continuously collected data from the radio and stored it in a dynamically growing buffer, and analyzed a fixed window length of data from the end of the buffer to make its classification. This process would then repeat, forming an online real-time brain-computer interface with MATLAB. Depending on the application that the BCI was used for, different actions were implemented to be performed upon particular stimuli classifications.

5. APPLICATIONS

For BCI-application interface development and testing, a computer game was utilized before interfacing the BCI with the automotive simulator.

5.1 Computer Game Interface

The interface with the computer game was accomplished by simulating keyboard presses that represented the classical user interface with the game (e.g. left key for “left”, x key for “jump”, etc.), which was achieved with the use of the Java Robot class. It was found that a delay of 50 ms was required between simulating key presses and key releases when interfacing with the computer game, and a delay of approximately 150 ms was required after a key release before successive key presses would be recognized by the game.

A maze-based computer game in which a character was able to move left, right, up, and down was interfaced with. Three BCI controls were successfully implemented as left (5.83 Hz), right (6.75 Hz), and up (8.65 Hz), using the high-power LEDs described in Section 3.1 mounted on the computer monitor. The BCI was tested using a section of the game where the down action was not required to navigate the maze: 16 steps through the maze in 7 directions were completed by the user in less than 30 seconds.

5.2 Driving Simulator Interface

The OKTAL advanced driving simulator installation at the Centre for Accident Research and Road Safety – Queensland utilizes SCANeR: a driving simulation software package used for vehicle ergonomics and advanced engineering studies, road traffic research and development, and studies on human factors in driving [5]. There are several independent modules in SCANeR which manage the hardware for the simulator sound, visualization, motion and feedback controls. The SCANeR system was modified to also run the authors’ custom-made executable module, which interfaced with the MATLAB-based BCI and overrode the appropriate signals received from the Data Acquisition module.

The structure of the simulator interface where input commands were received from both the driving controls and the BCI is shown in Figure 2. The BCI was set up to continuously output the appropriate control signals according to the classification of the user’s intent. During simulation, the custom SCANeR module would invoke the MATLAB-based BCI to run, cancel out particular traditional driving controller signals, and replace these signals with the control signals generated by the BCI.

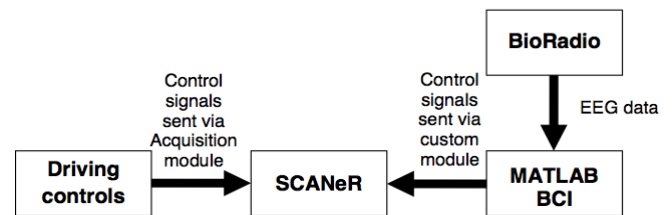


Figure 2. Diagram of the BCI integration with the SCANeR driving simulation software.

The need for cancelling driving controller signals arose when the brain-computer interface was sending control signals for the same function that the controller was. For example, if the car steering wheel was enabled while the BCI was trying to simulate the steering wheel control, they would interfere with each other and neither will control successfully. One option was to completely disable the car driving controls, but as not every function would be controlled by the BCI, the functionality of the driving simulator would have been drastically reduced. For this reason, the individual control signals were cancelled for the functions that the BCI was controlling.

The SSVEP light stimuli were mounted in the user’s view on the dashboard. The three stimuli implemented were 5.83 Hz, 6.75 Hz, and 8.65 Hz. The BCI classifications of these stimuli frequencies were able to be linked with any controls accessible by the driving simulation software. For example, the commands could be linked to acceleration, deceleration and cruising, with the steering and other car controls operated as usual by the user (Figure 3). Alternatively, the commands could be linked to steer left, steer right and steer straight, with the acceleration and other car controls operated by the user. After approximately 10 minutes of training in this configuration the user was able to keep the car on the road and complete turns around corners using the BCI control: this was application-specific training and no user training, only user-specific tuning as described in Section 3.4, was required for the unconsciously generated SSVEP response.

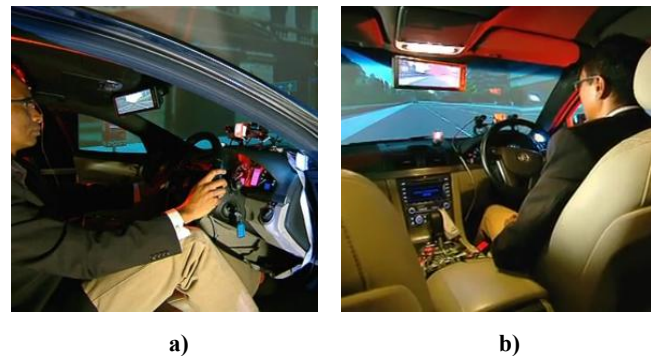


Figure 3. BCI control of the a) acceleration and b) steering of the CARRS-Q advanced driving simulator.

6. CONCLUSION

The methodology and implementation of a non-invasive electroencephalography-based brain-computer interface to control functions of a car in a driving simulator has been described.

An assessment of potential BCI paradigms was performed, leading to the decision to use steady-state visual-evoked potential (SSVEP) as the stimulus, due to its requirement of only

intermittent visual attention [3] and lack of user training and mental focus required [4].

In this stimulation paradigm, the user directs their visual attention to one of numerous uniquely frequency-modulated lights/images that represents their decision [8]. Electronic stimuli in the form of lights were chosen as the method of eliciting SSVEP, as they do not require a display/monitor and were more appropriate for distribution throughout a car.

The SSVEP paradigm was successfully implemented using high-power LEDs and the electrode placement of Oz-Fz with Fz as ground. Matched filter detection was implemented to determine the most significant stimulus frequency in the user's brain response and therefore decode the user's intent [2], which then translated to a particular action based on the application of the BCI.

A three-class online brain-computer interface was implemented in MATLAB and interfaced with a computer game utilizing simulated keystrokes through the Java Robot class.

The BCI was also successfully interfaced with the SCANer advanced automotive simulator to control the steering of the car. The communication with SCANer was achieved by developing a custom executable module to cancel the appropriate control signals obtained from the traditional car controls, communicate with the MATLAB BCI to determine the user's intent, and write the appropriate control signals to the car. A three-class BCI was implemented with left, right, and straight steering as the actions, resulting in the user successfully being able to control the position of the car on the road with their brain signals.

The accuracy and reliability of BCIs require significant improvements before being deployed in safety-critical applications such as driving, taking individual neurological differences, driver distractions, mental workload and musculoskeletal movement artifact robustness requirements that arise in this context into consideration. The research outlined in this paper of a brain-computer interface with a safe, simulated driving environment such as that at CARRS-Q will be imperative

in overcoming these challenges and making brain-controlled cars safe for on-road use.

7. REFERENCES

- [1] Centre for Accident Research and Road Safety – Queensland (CARRS-Q) Advanced Driving Simulator.
<http://www.carrsq.qut.edu.au/simulator/>
- [2] Chang, Hsiang-Chih, Hua-Ting Deng, Po-Lei Lee, Chi-Hsun Wu and Kuo-Kai Shyu. 2010. Real-Time Control of an SSVEP-Actuated Remote-Controlled Car. In *Proceedings of SICE Annual Conference 2010, August 2010*.
- [3] Kelly, Simon, Edmund Lalor, Richard Reilly and John Foxe. 2005. Visual Spatial Attention Tracking Using High-Density SSVEP Data for Independent Brain-Computer Communication. *IEEE Transactions on Neural Systems and Rehabilitation Engineering* 13:2(172-178).
- [4] Lotte, Fabien. 2008. Study of Electroencephalographic Signal Processing and Classification Techniques towards the use of Brain-Computer Interfaces in Virtual Reality Applications. PhD diss., National Institute of Applied Sciences (INSA) Rennes.
- [5] OKTAL. 2011. *User Manual: SCANerTM Studio Version 1.1*. France: OKTAL Publication.
- [6] Tan, Desney, Anton Nijholt. 2010. Brain-Computer Interfaces and Human-Computer Interaction. In *Brain-Computer Interfaces*, edited by Desney S. Tan and Anton Nijholt, 3-19. London: Springer-Verlag.
- [7] Wang, Hongtao, Ting Li, Zhenfeng Huang. 2010. Remote Control of an Electrical Car with SSVEP-Based BCI. In *IEEE International Conference on Information Theory and Information Security, January 2010*.
- [8] Zhu, Danhua, Jordi Bieger, Gary G. Molina, Ronald M. Aarts. 2010. A Survey of Stimulation Methods Used in SSVEP-Based BCIs. *Computational Intelligence and Neuroscience* 2010.